PROJECT REPORT

Auto Ticket Classification using Flow Designer

ServiceNow | School IT Helpdesk Automation

Team ID	SWTID-2026-8307
Project Domain	IT Service Management / Workflow Automation
Platform	ServiceNow Flow Designer
Document	Final Project Report

Prepared for academic project submission

Note: This report describes the proposed workflow and test plan. Actual execution results and screenshots should be added after implementation and testing.

PROJECT REPORT FORMAT

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1. INTRODUCTION

Schools depend on computers, projectors, Wi-Fi and online accounts for daily teaching and administration. When these services fail, students and teachers contact the IT helpdesk. Each incident must be reviewed and assigned to an appropriate category before it can be routed and resolved.

This project proposes a keyword-based automation workflow in ServiceNow Flow Designer. It examines the Short Description and Description fields of a newly created incident and assigns a category based on configured keyword rules.

1.1 Project Overview

The project focuses on reducing manual effort in the initial classification of school IT incidents. The workflow uses rule-based matching rather than a trained machine-learning model. The main categories and example keywords are shown below.

Keywords found in incident text	Assigned category
Wi-Fi, network, internet	Network
Projector, display, screen	Hardware / AV
Password, login, account	Access / Password
Slow, hanging, performance	Performance / Computer

1.2 Purpose

• Reduce the time spent manually reviewing and categorizing incidents.

• Apply consistent category rules to common helpdesk requests.

• Support faster triage by assigning a relevant category when an incident is created.

• Provide a clear, repeatable workflow that can be tested and maintained.

2. IDEATION PHASE

The ideation phase identifies the recurring helpdesk problem and outlines a practical automation approach.

2.1 Problem Statement

I am a school IT helpdesk staff member trying to classify incoming incident tickets quickly. However, I must read each ticket and select its category manually. Because requests arrive with different wording and are handled by different staff members, classification can take time and become inconsistent. This makes ticket triage less efficient.

PS-1: School IT staff spend time manually categorizing Wi-Fi, projector, login and computer-performance incidents.

PS-2: Students and teachers may experience delays in ticket routing when incident categories are assigned inconsistently.

2.2 Empathy Map Canvas

Says	Thinks	Does	Feels
“My issue has been reported.”	“Will my ticket reach the right team?”	Reads ticket details and selects a category.	Busy when many tickets arrive; frustrated by repeated manual work.

2.3 Brainstorming

• Use consistent keyword rules for common incident types.

• Trigger the classification flow when a new incident is created.

• Evaluate both Short Description and Description fields.

• Define a safe fallback for tickets with no matching keyword.

• Test each category with representative sample incidents.

3. REQUIREMENT ANALYSIS

This section lists the user journey, functional requirements, data movement and technologies needed for the proposed workflow.

3.1 Customer Journey Map

Stage	User / system activity	Opportunity
1. Report issue	Student or teacher submits an incident.	Make incident entry straightforward.
2. Incident created	ServiceNow stores the incident details.	Capture useful text in both description fields.
3. Classification	Flow checks configured keywords.	Apply consistent rules.
4. Category assigned	Flow updates the incident category.	Reduce manual categorization.
5. Follow-up	Helpdesk reviews and handles the ticket.	Verify category and route as needed.

3.2 Solution Requirement

• A ServiceNow Incident record must contain Short Description and Description fields.

• A Flow Designer flow must be configured to trigger on incident creation.

• The flow must evaluate the text against the agreed keyword rules.

• The flow must assign the mapped category when a rule matches.

• A default or review path should be defined for unmatched or ambiguous incidents.

• Execution outcomes should be verifiable through incident records and flow execution details.

3.3 Data Flow Diagram

Step	Input / Process / Output
Input	Student/teacher submits incident details.
Process 1	ServiceNow creates the Incident record.
Process 2	Flow Designer reads Short Description and Description.
Process 3	Keyword conditions identify the matching incident type.
Output	Relevant category is written to the incident record.
Exception path	No match → configured default or manual review.

3.4 Technology Stack

Technology / component	Purpose
ServiceNow	IT service management platform and incident records.

Technology / component	Purpose
Flow Designer	Trigger-based workflow automation.
Incident table	Stores incident details and category.
Keyword conditions	Rule-based matching of incident text.
Test incident records	Validate category assignment and consistency.

4. PROJECT DESIGN

The design uses a simple rule-based workflow. It is intended to classify common incidents automatically while leaving unclear cases available for review.

4.1 Problem Solution Fit

Problem	Design response
Manual category selection takes time.	Automate category assignment through Flow Designer.
Similar issues may be categorized differently.	Use documented keyword-to-category rules.
Ticket text may not match a rule.	Provide an unmatched/default handling path.

4.2 Proposed Solution

When an incident is created, the flow reads its Short Description and Description. It evaluates the text against the configured keyword groups. If a group matches, the flow sets the corresponding category. If none matches, the flow follows the configured fallback behavior. The helpdesk can review the resulting incident and correct exceptional cases if required.

4.3 Solution Architecture

Order	Component	Responsibility
1	Student / Teacher	Submits the incident.
2	ServiceNow Incident	Stores the incident fields.
3	Flow Designer Trigger	Starts the flow on incident creation.
4	Keyword Conditions	Checks text for configured terms.
5	Category Update	Writes the selected category.
6	Helpdesk Review	Handles unmatched or exceptional tickets.

5. PROJECT PLANNING & SCHEDULING

The following is a suggested implementation sequence. Update the schedule to reflect the team's actual work dates.

5.1 Project Planning

Phase	Task	Deliverable
1	Understand the helpdesk problem	Problem statement and scope
2	Define categories and keywords	Keyword mapping table
3	Configure incident trigger	Flow trigger setup

Phase	Task	Deliverable
4	Build classification conditions	Keyword decision logic
5	Configure category updates	Incident field update
6	Test matching and fallback cases	Test evidence and results
7	Document and review	Final report and screenshots

6. FUNCTIONAL AND PERFORMANCE TESTING

Testing should confirm that the flow triggers correctly, matches the intended keywords, updates the correct category and handles unmatched incidents without failure. Record actual outcomes after running the tests in the ServiceNow instance.

6.1 Performance Testing

Test ID	Scenario	Expected result	Actual result
FT-01	Create incident with Wi-Fi/network keyword.	Category becomes Network.	To be recorded
FT-02	Create incident with projector/display keyword.	Category becomes Hardware / AV.	To be recorded
FT-03	Create incident with password/login keyword.	Category becomes Access / Password.	To be recorded
FT-04	Create incident with slow/hanging keyword.	Category becomes Performance / Computer.	To be recorded
FT-05	Create incident with no configured keyword.	Configured fallback/review path is followed.	To be recorded
PT-01	Measure time from incident creation to category update.	Record observed processing time; compare with project target.	To be recorded
PT-02	Create several incidents close together.	Flow processes records consistently; note errors or delays.	To be recorded

Do not report accuracy percentages or response-time claims until they have been measured. For a rule-based workflow, classification accuracy can be calculated as correctly categorized test incidents divided by total test incidents, multiplied by 100.

7. RESULTS

The expected outcome is automatic category assignment for incidents that contain configured keywords. The final results section should be completed using evidence from the implemented ServiceNow flow.

7.1 Output Screenshots

Screenshot ID	Evidence to insert	Observation
SS-01	Flow Designer trigger and workflow configuration	Add after implementation
SS-02	Incident with Wi-Fi/network text and assigned category	Add after execution
SS-03	Incident with projector/display text and assigned category	Add after execution
SS-04	Incident with password/login text and assigned category	Add after execution
SS-05	Incident with slow/performance text and assigned category	Add after execution

Screenshot ID	Evidence to insert	Observation
SS-06	Flow execution details or unmatched-case handling	Add after execution

8. ADVANTAGES & DISADVANTAGES

Advantages

• Reduces repetitive manual classification for common incident types.

• Applies the same documented rules to each incident.

• Uses a visual low-code workflow within ServiceNow.

• Can be extended by adding or refining keyword rules.

Disadvantages / limitations

• Keyword matching may miss issues described with unexpected wording or spelling.

• A ticket containing keywords from multiple groups may need priority or conflict-handling logic.

• The workflow depends on correct configuration and maintenance of rules.

• Unmatched incidents may still require manual review.

9. CONCLUSION

Auto Ticket Classification using Flow Designer proposes a practical way to automate the first stage of school IT helpdesk incident handling. By checking the Short Description and Description fields against keyword rules, ServiceNow can assign categories for common Wi-Fi, audiovisual, account-access and computer-performance issues. The effectiveness of the implemented workflow should be confirmed through functional tests, performance observations and screenshots.

10. FUTURE SCOPE

• Add more incident categories and maintain a centrally reviewed keyword list.

• Introduce rule priority for incidents containing keywords from multiple categories.

• Add synonym and spelling-variation handling.

• Route categorized incidents to the appropriate support group.

• Use reporting dashboards to monitor category trends and classification corrections.

• Explore NLP or machine-learning classification if sufficient labeled incident data becomes available.

11. APPENDIX

Source Code (if any)

ServiceNow Flow Designer configuration is the primary implementation artifact. Add exported flow details, scripts (if used), or configuration documentation here. Do not include credentials, tokens or other secrets.

Dataset Link

Not applicable to the current rule-based design unless a sample incident dataset is used. If test data is maintained externally, add its approved link here.

GitHub & Project Demo Link

GitHub Repository: [Add repository URL]
Project Demo: [Add demo URL, if available]

Reference structure followed: Project Report Format template supplied by the project team.